North Branch of the Potomac River Tailwater Trout Fisheries

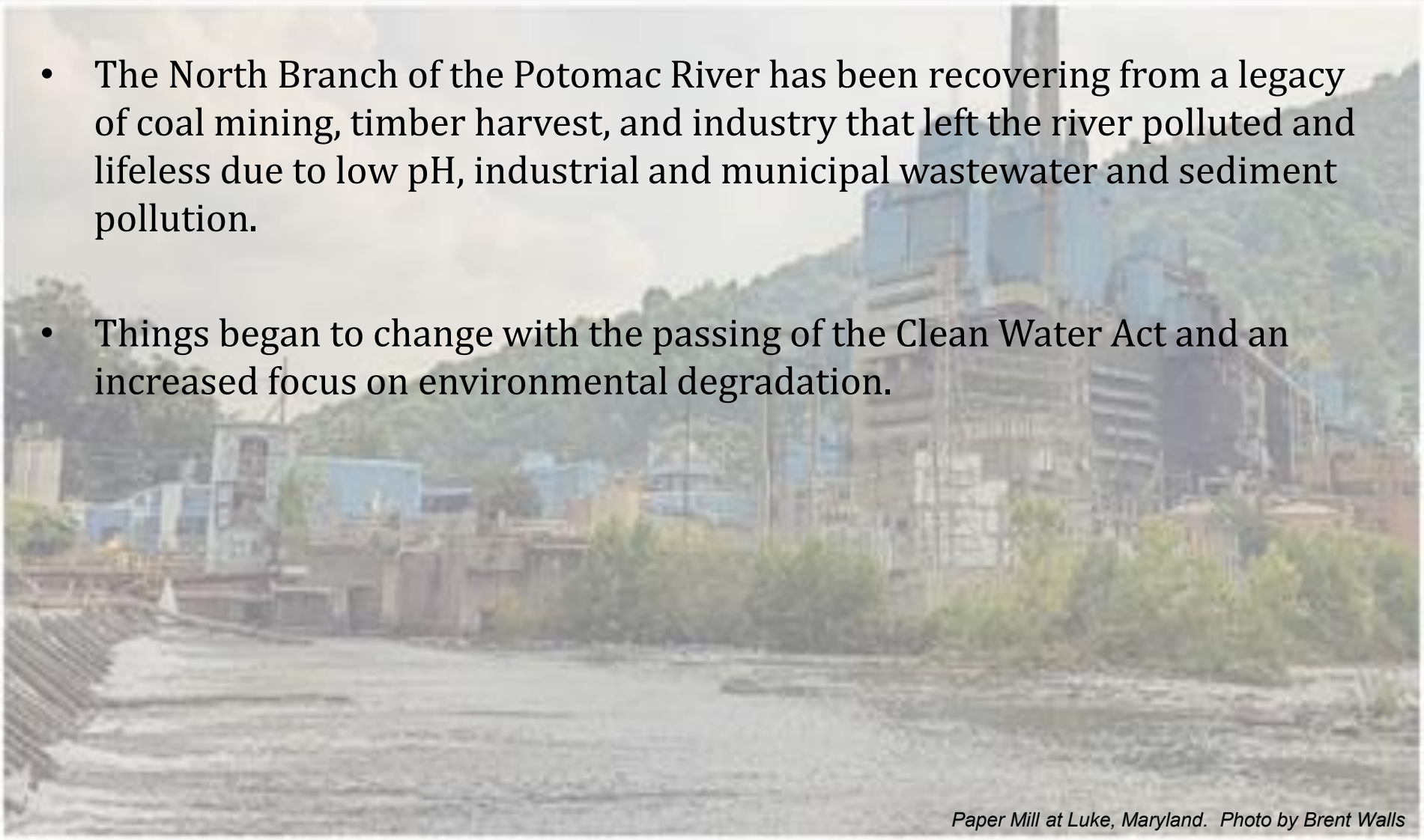
USACE Environmental Flows Workshop

August 6, 2025

History of the Fishery

- The North Branch of the Potomac River has been recovering from a legacy of coal mining, timber harvest, and industry that left the river polluted and lifeless due to low pH, industrial and municipal wastewater and sediment pollution.

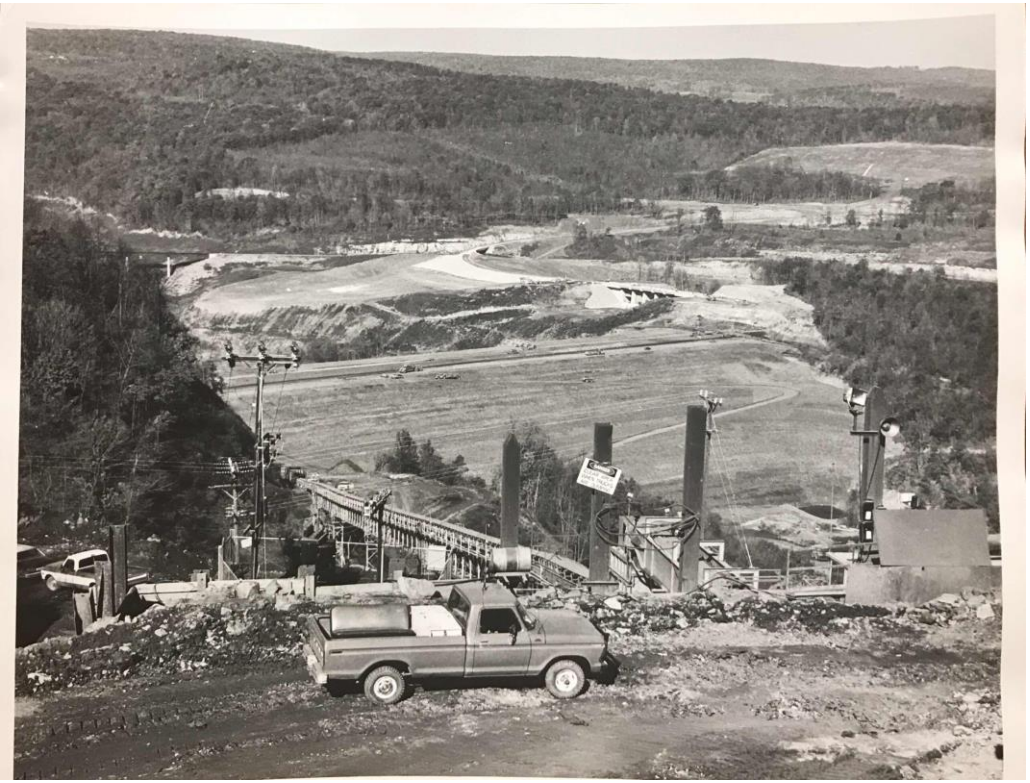
- Things began to change with the passing of the Clean Water Act and an increased focus on environmental degradation.



Paper Mill at Luke, Maryland. Photo by Brent Walls

History of the Fishery

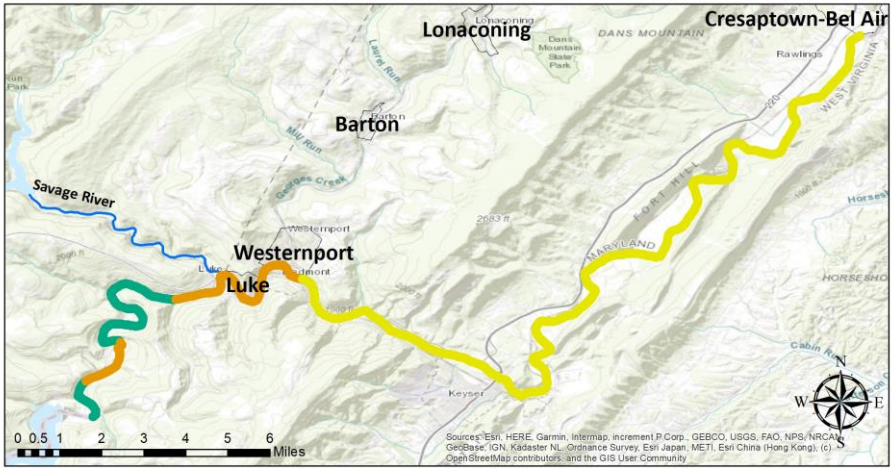
- Jennings Randolph Reservoir (formerly Bloomington Dam) was completed in 1981 and brought online in 1982.
- JRR provided benefits to downstream water quality and trout were successfully reproducing in the tailwater by the early 1990s.
- Since then, a world-class trout fishery has developed in the tailwater of JRR, sufficient to support more than 10 full time trout fishing guides



Construction of the Bloomington Dam. Photo credit: The Baltimore Sun

Trout Management in the North Branch Tailwater

- Nearly 30 miles of the NBPR downstream of Jennings-Randolph Reservoir is now occupied by trout
- Includes wild trout, Put and Take, and Put and Grow management areas with special fishing regulations in some areas
  - Wild Trout: self-sustaining fisheries due to natural reproduction
  - Put and Take: areas stocked with adult trout annually with the expectation of harvest
  - Put and Grow: areas stocked with fingerling trout to support little/no natural reproduction but is capable of year-round trout survival

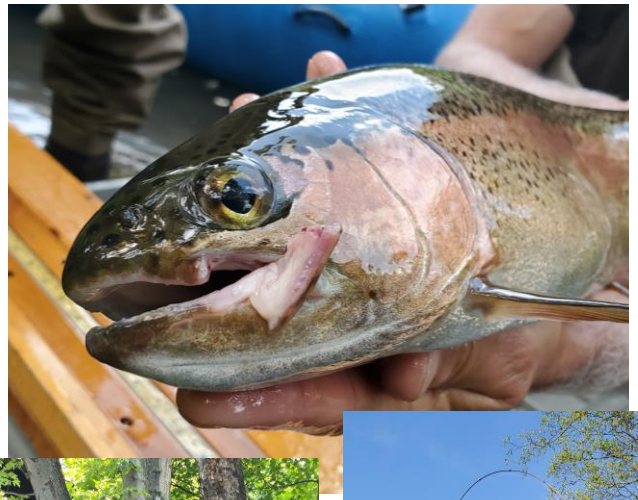


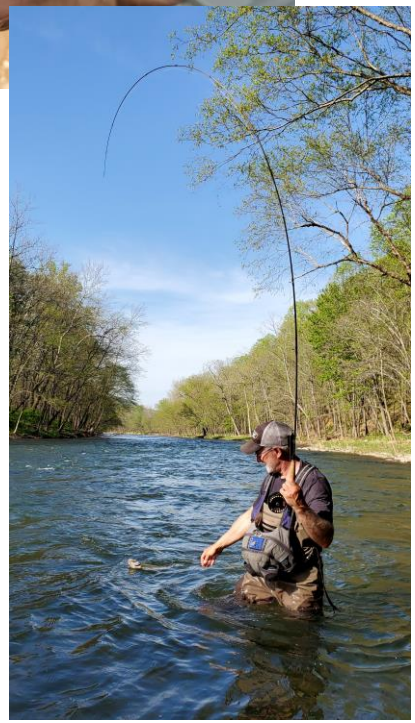
CRAT Management Area
Put and Take Management Area
Catch and Release Management Area

North Branch of the Potomac River
Special Trout Management Areas

Trout Management in the North Branch Tailwater

- The resource is managed using data collected through both fishery dependent and independent surveys.
- Electrofishing surveys are conducted annually by MD DNR Freshwater Fisheries and Hatcheries staff to monitor the fishery between Westernport and Black Oak. Surveys are also conducted as flows allow between Barnum and Bloomington.
- An ongoing PIT tagging study is providing information about age, growth, mortality and movement of the trout resource.
- A recent creel survey in the Catch and Release, All Tackle Management Area (CRAT) estimates that every adult trout in the river is caught on average more than twice a year!
- Temperature is monitored annually using a combination of temperature loggers and a real-time temperature probe.



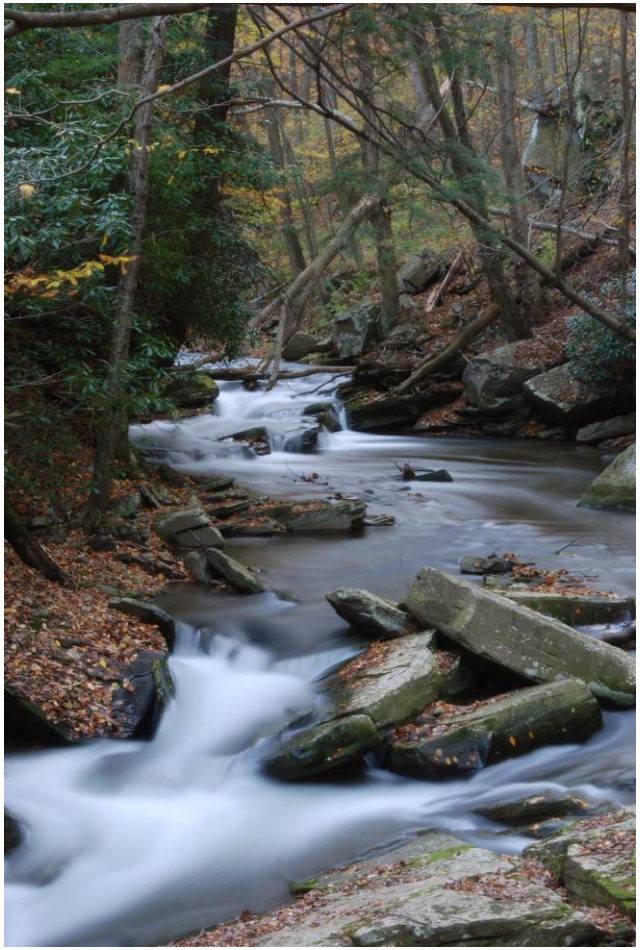


Water Quality Requirements

MARYLAND DEPARTMENT OF NATURAL RESOURCES

Temperature

- **Primary importance is cold water year-round**.
- In general, trout prefer water temperatures <20˚C (68˚F)
- Optimal temperature requirements for good growth and survival of brown trout are 12°C to 19°C (Mills 1971; Brown 1973;Frost and Brown 1967,1975), with a temperature tolerance range of 0°C to 27°C (Maciolek and Needham 1952; Mills 1971).
- Streams designated as Use Class III (or III-P) waters (nontidal cold water use) in Maryland provide the most stringent temperature criteria, requiring temperatures to stay below 20°C.
- Use Class IV (or IV-P) designations (recreational trout-stocking waters) also represent cooler water resources and have a water temperature criterion of below 23.9°C.

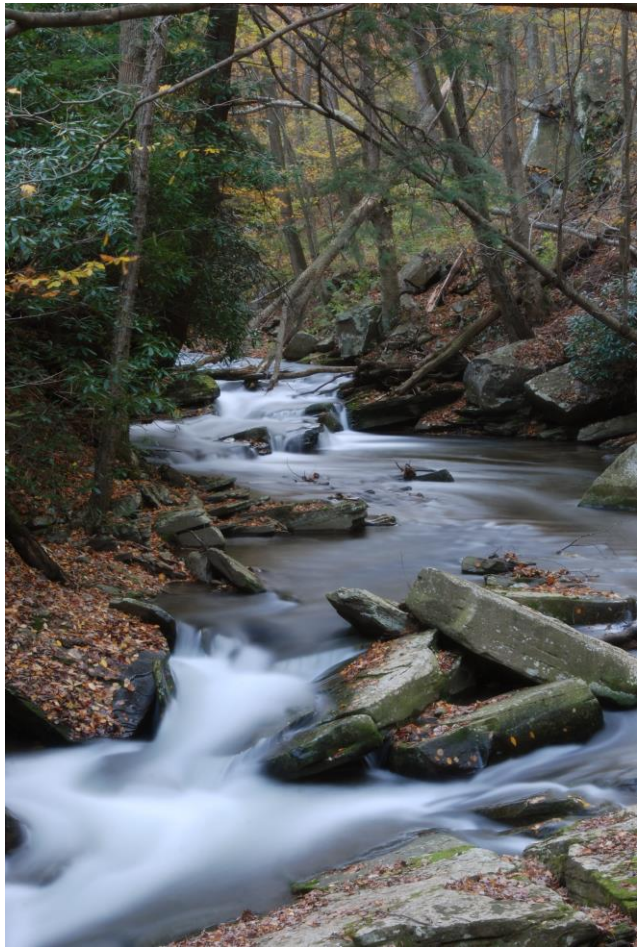


Water Quality Requirements

MARYLAND DEPARTMENT OF NATURAL RESOURCES

Dissolved Oxygen

- Generally, high dissolved oxygen levels (>8 parts per million)
- In the summer (> 10 C), trout generally avoid water with dissolved oxygen levels of less than 5 mg/l.
- Insufficient dissolved, oxygen is typically only a problem in streams with a slow current, excessive temperature, and a high biological oxygen demand.
- Use III Waters - The dissolved oxygen concentration may not be less than 5 milligrams/liter at any time, with a minimum daily average of not less than 6 milligrams/liter.



Water Quality Requirements

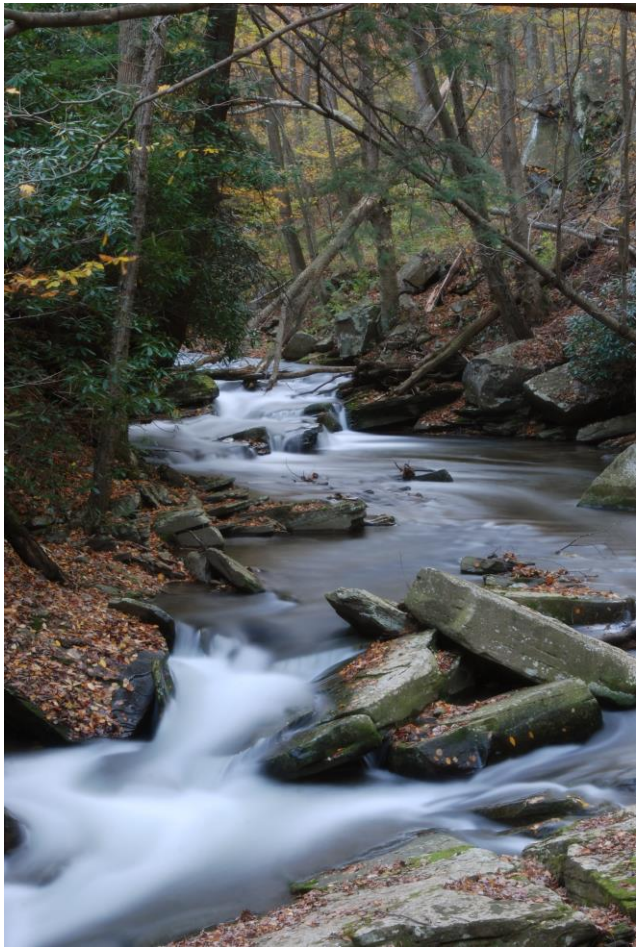
pH

- Generally a pH ranging from 5.5 to 8.5 (pH of 7 is neutral);

  Brook trout - pH from 5.5 - 8.0, the most acid tolerant trout species;

  Brown trout - pH from 6.5 - 8.5, prefer neutral to slightly alkaline waters;

  Rainbow trout - pH from 6.5 – 8.5, prefer more alkaline waters



Do We Need Change?

The current operational guidelines at JRR are adequate to provide necessary water quality to support the tailwater trout fishery. What would a change in flow regime do to help???

The development of Environmental Flows and adaptive management process:

1. **May be what is needed to allow natural reproduction of trout to improve in the river;**
2. **Improve overall habitat;**
3. **Affect growth rates;**
4. **Increase survival of young-of-year trout;**
5. **Mimic a more natural flow regime that trout have evolved with.**

Transitioning to wild trout…

Trout Reproduction

- Brook trout & brown trout spawn in the fall – mid-October to mid-November; rainbow trout spawn late winter - early spring, as early as mid-February
- Spawning areas are in cobble/gravel (marble to pea size) with upwelling flows and/or in shallow riffle areas with constant flow
- Constant flow keeps eggs oxygenated and clean of sediment - trout eggs are highly susceptible to mortality from sedimentation
- Spawning periods, egg incubation, and fry life stages are extremely sensitive and critical periods

Brook trout establishing a redd in a gravel area with adequate flow

Trout Reproduction

- Eggs are deposited and fertilized in a redd (nest) built by the female, then buried 1”- 2” deep in gravel
- Eggs develop in the redd for a period of several months and hatching typically occurs during early spring (brook and brown trout) or late spring (rainbow trout)
- Fry emerge from the redd with a yolk sac used to sustain them until they transition to a diet of zooplankton
- The yolk sac is eventually absorbed as the fry transition to the free swimming fingerling life stage

The Maryland Department of the Environment Use III (non-tidal coldwater) stream closure period (October 1 - April 30) was established to protect these critical life stages: COMAR 26.17.04.11

Developing eggs

Fry with yolk sac

Fingerling stage

What Trout Need… Flow

Seasonal Flow Patterns

Fall: increasing flows concurrent with decreasing photoperiod trigger spawning activity and provide access to suitable spawning locations during the fall (October-December) for brook and brown trout and late winter/early spring (December-February) for rainbow trout.

Winter: adequate base flows to oxygenate developing embryos and to prevent redds from freezing

Spring: stable flow conditions without abnormally high or low flows to protect newly emerging alevins/fry

Summer: adequate flows to maintain coldwater temperature regime to support year-round trout survival and provide appropriate habitat for trout of all life stages

Annually…

…an ***annual base flow ~50% of the average annual daily flow*** is considered excellent for salmonid production, a base flow of 25% to 50% is considered fair to good, and one of ~25% is considered poor (adapted from Tennent 1976; Binns and Eiserman 1979; Wesche 1980).

…Nehring and Anderson (1982, 1983) ***consider a peak flow of about five times the magnitude of an excellent base flow, or about two times the average annual daily flow*** (Lister and Walker 1966) to be acceptable for good salmonid production and ***peak flows approaching twice these limits are considered progressively more destructive***.

Critical Metrics

1. Summer Flows: Adequate summer flows maintain habitat during dry seasons, ensure cool water temperatures, and prevent habitat fragmentation.

Ideal Conditions:

- Flow sufficient to maintain **depths of ≥20–30 cm** in riffles and pools.
- Water temperatures typically **<20°C (68°F)**, ideally **<17°C (63°F)** for coldwater species.
- Avoid flow drops that isolate pools or reduce oxygen.

2. Peak Flows (Spring Snowmelt / Storm Events): Peak flows help to mobilize sediment, maintain channel structure, connect floodplains, and signal migration or spawning efforts.

Ideal Conditions:

- Annual peaks that mimic natural magnitude and duration.
- Not too flashy (rapid rise/fall) — avoid scouring eggs or injuring fry/juveniles.
- Frequent enough to maintain habitat complexity, but not so high they cause erosion.

Critical Metrics

<u>3. Spawning Flows (Fall for brook and brown trout; spring for rainbow trout)</u>: Appropriate flows during spawning periods provide access to spawning gravels, clean substrate, and proper depth and velocity for redd construction.

<u>Ideal Conditions:</u>

- Flows that ***submerge gravel beds in 15–40 cm of water.***
- Velocities of ***0.3–0.9 m/s (1–3 ft/s) at spawning sites***.
- Minimal sedimentation and stable elevated base flows (25-50+% of average annual daily flows) to provide adequate aeration of eggs and avoiding desiccation or scouring.
- Waters (1976) set the optimal water depth for brown trout redd construction at 24.4 to 45.7 cm, with a suitable range of 12.2 to 91.4 cm, whereas Shirvell and Dungey (1983) reported that 31.7 cm was the preferred spawning depth for brown trout.
- Optimal spawning gravel conditions for brown trout are assumed to be < 5% fines; > 30% fines are assumed to result in low survival of embryos and emerging fry.

Critical Metrics

4. Incubation and Emergence Flows: Appropriate flows to protect eggs and alevins in redds and maintain dissolved oxygen.

Ideal Conditions:

- Avoid dewatering redds (critical in winter).
- Stable, moderate flows without sudden drops or spikes.

5. Rearing Flows (Juvenile Growth): Flows that provide food, cover, and refuge areas for juvenile trout.

Ideal Conditions:

- Diverse habitats (riffles, pools, woody debris).
- Flows that support habitat complexity (e.g., pocket water, undercut banks).
- Flow velocities that allow drift feeding: ***0.2–0.6 m/s are optimal*** for juveniles.

In Summary…

Current dam operational guidelines have maintained suitable temperatures for trout in the tailwater downstream as far as Pinto, Maryland.

Adult trout densities have averaged more than 350 adult trout per mile in recent years, with asought-after trophy component to the fishery.

Rainbow trout reproduction in the upper portion of the tailwater (JRL to Bloomington) has been sufficient to support the popular fishery. Rainbow trout reproduction in the lower reaches of the tailwater is suspected, but undocumented.

Brown trout reproduction has been documented in the lower tailwater (Westernport to Pinto) but the contribution of those wild fish to the overall fishery is currently unknown.

MD DNR Freshwater Fisheries and Hatcheries Division, in cooperation with the WV Division of Natural Resources continues to maintain the fishery through annual stockings of adult hatchery-reared trout as well as over 100,000 hatchery-reared fingerling trout.

Continued cooperation with the USACE will be critical to maintain the popular trout fishery downstream of JRL.